Fair Test Planner: Change One Thing

I can plan a safe fair comparison by naming one changed variable, one measured result, the controls and the repeats.

CHANGE
one planned difference

MEASURE
one result with a unit

CONTROL + REPEAT
same method, checked evidence

[Open the original interactive lesson]

What success looks like

- Write or select one clear question for the W14 rock investigation.
- Name what changes, what is measured and at least three important controls.
- Prepare a repeat plan, blank table and safety stop for a method pilot.

Retrieve and reconnect

- Last lesson: what method question did your audience bank?
- Last lesson: why did irrelevant evidence not help the choice?
- Last week: name one condition held steady in the W11 fixed water method.
- Lead-in: explain why a fair-test question must name what will be compared.

Team A gives Sample A 5 mL for 120 seconds. Team B gives Sample B 10 mL for 60 seconds. Could a final comparison show which sample let more water through?

Think first. Point, say, sign or write your choice. Give one reason.

A. Yes. Any two collected-water numbers make a fair comparison.

B. Not yet. Sample, starting volume and time all changed, so the cause is unclear.

C. Yes, because using more water always makes a better test.

Look closely · what is the evidence?

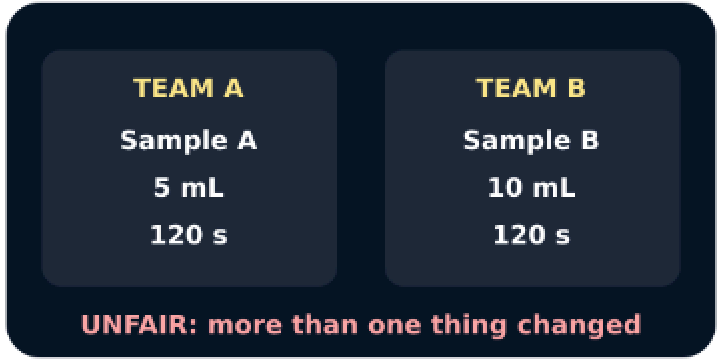
Planned difference: the supplied sample code should change.

Unplanned difference: starting water changes from 5 mL to 10 mL.

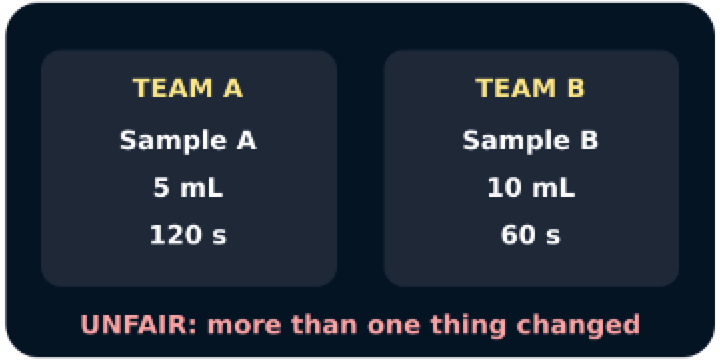
Unplanned difference: test time changes from 120 seconds to 60 seconds.

Look closely · what is the evidence?

Two teams differing only in sample



Team B used more water



Team B also used less time

Words we will use

- variable — a factor that can change
- control — keep this the same
- repeat — do the same method again

Watch the model

- Write the W14 question: Which supplied coded sample lets more water through in 120 seconds under this method?
- Change only the sample code. Measure water collected below in mL. Keep 5 mL starting volume and 120-second time the same.
- Add the less visible controls: same sealed collar area, matched prepared thickness, dry starting condition, stable rig and measuring tool.

Which plan changes one factor and measures one result?

A. Change sample code; measure water collected in mL; keep volume, time, area and thickness the same.

B. Change sample, water volume and time; measure whichever result appears.

C. Scratch A, wet B and compare the two different observations as permeability.

Commit to an answer. Show the evidence you used. Listen, then revise if needed.

Build the explanation

- Plan the W14 repeats before seeing final results. Use staff-prepared dry matched pieces and preserve every valid raw value.
- Prepare the pilot question separately: Can our instructions, seal, timer and table produce one valid practice record? The W13B pilot tests the method, not which final sample wins.
- Add the safety stop: staff build and pretest rigs; stop for leaks, instability, damage, flaking, dust or unclear measurement and switch to the fixed pilot route.

Find and repair the misconception

Planner glitch: a fair test is one where all samples get the same result.

A. Keep it: equal outcomes prove fairness.

B. Repair it: fair describes method control while one planned factor changes; final results may differ.

C. Repair it: use the pilot value as the final conclusion before testing A and B.

W14 fair-test control board

Route each plan card to CHANGE, MEASURE, KEEP THE SAME or REPEAT / PILOT / SAFETY.

- Read the factor or action.
- Decide its job in the plan.
- Commit and repair any mismatch.
- Freeze a complete method, then rehearse it for approval.

Use the numbered cards in your pupil pack. Take turns to choose, justify and check. You may direct a partner to move a card.

Choose a group · defend your placement

- CHANGE
- MEASURE
- KEEP THE SAME
- REPEAT / PILOT / SAFETY

Rehearse: Change __. Measure __. Keep __, __ and __ the same. Pilot __. Repeat __ in W14. Stop if __.

Your independent mission

Produce a W14-ready fair-test plan and a separate W13B pilot checklist, leaving all final sample result cells blank.

Record: W14 question, prediction, changed variable, measure with unit, at least three controls, repeat plan, blank results table, W13B pilot checks and safety stop.

Choose your support and challenge

- Supported: Build the method from fixed picture cards and choose one pilot check and one safety stop.
- Standard: Write the full plan, justify three controls and prepare separate pilot and final-data sheets.
- Stretch: Evaluate a flawed alternative, justify thickness and exposed-area controls, and explain what the pilot can and cannot establish.

Show what you know

Explain why W13B tests the method but does not answer the final A/B question.

Point to, read or explain the evidence you made. Choose one next step after the adult has listened.